SPOORTHI M

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Education

Amrita School of Computing, Amrita Vishwa Vidyapeetham **Sep. 2021 – May 2025**

Bachelor of Technology in Computer Science & Engineering *Bengaluru, India*

CGPA: 9.09/10 | University Rank: 6th out of 200 students

Research Interests

Medical Image Analysis | Neural Radiance Fields (NeRF) | Computational Pathology | 3D Reconstruction & Volumetry | Deep Learning | Computer Vision | Healthcare AI

Publications

Sparse-View Lung Nodule Volumetry via AReT: Anatomy-Regularized TensoRF | *Under Review* **2026**

arXiv:2606.02639

- M Spoorthi, P Suja. Manuscript under review.
- Identified and resolved an unreported TensoRF failure mode — default density shift $\Delta = -10$ suppresses all density gradients via dead ReLU throughout training — by setting $\Delta = 0$, restoring gradient flow with zero architectural changes; enables sparse-view X-ray attenuation reconstruction in TensoRF for the first time with no modifications to the base architecture.
- Proposed AReT, an anatomy-regularized TensoRF framework for 3D lung nodule volumetry from only three orthogonal digitally reconstructed radiographs (coronal, sagittal, axial); achieved Pearson $r = 0.983$, Spearman $\rho = 0.943$ ($p < 0.0001$), median APE 11.4%, and Bland–Altman bias -77.3 mm^3 on clinically actionable nodules ≥ 10 mm ($n = 14$), an 8.4× improvement over spherical approximation.
- Designed thorax-specific $\ell_1 + \mathrm{TV}$ regularization exploiting air-dominant thoracic anatomy as a physics-informed inductive bias, enabling stable 3D reconstruction from 49,152 training rays with only 869,555 parameters across 3,000 training iterations; implemented Beer–Lambert DRR simulation and HU-calibrated DICOM preprocessing pipeline end-to-end in PyTorch on NVIDIA T4 (≈8 min/patient).
- Conducted a systematic 11-strategy progressive ablation across MedNeRF variants, CheXNet, MONAI UNet, and TensoRF configurations on LIDC-IDRI ($n = 19$), demonstrating anatomy-regularized prior-free TensoRF outperforms all generative-prior-guided approaches (Wilcoxon $W = 0$, $p < 0.0001$).

AMN: An Adaptive Multi-Scale Fusion Network with Boundary and Uncertainty Modeling for Nuclei Segmentation | *Accepted at ICETCI 2026* **2026**

arXiv:2606.07633

- M Spoorthi, P Suja. Manuscript under review.
- Proposed AMN, a dual-encoder segmentation framework pairing a Swin Transformer and ResNet-50 feature pyramid, fused at four spatial scales via a learned per-channel sigmoid gate $\alpha \in [0,1]^{256}$, dynamically blending global context and local texture for seven-class nuclei segmentation in H&E histopathology images.
- Designed a multi-task prediction head producing simultaneous binary segmentation, boundary detection, seven-class classification, and heteroscedastic uncertainty outputs, trained with a composite loss of class-weighted focal loss (γ=2.0), boundary-aware BCE (positive weight 10×), and uncertainty-modulated cross-entropy; achieved mean F1 of 0.6869 and $\mathrm{F1}_{\mathrm{Ly}}$ of 0.8187 on CoNIC (+12.6 points over ResU-Net).

- Benchmarked against six reimplemented baselines (U-Net, ResU-Net, DeepLabV3+, SegNet, ViT-Small, StarDist) on CoNIC (4,981 patches, 7 classes) with zero-shot cross-dataset evaluation on MoNuSeg (14 images, 7 organs), achieving highest binary F1 without retraining.
- Addressed severe class imbalance (background 83.87%, rare classes <1%) via per-class focal weights, weighted random sampling, and boundary-aware supervision; trained with AdamW, cosine annealing, 5-epoch linear warmup, and gradient accumulation (effective batch 64) for up to 200 epochs with early stopping.

Advanced Topic Modeling Techniques for Categorizing Software Vulnerabilities | *Accepted at ICCCNT 2025* **2025**

- U Tiwari, M Spoorthi, S Anirudh, N Prabhakar T.V. Accepted at the 16th IEEE International Conference on Computing, Communication and Networking Technologies (ICCCNT 2025), IIT Indore.
- Benchmarked five advanced topic modeling approaches — BERTopic (four configurations: UMAP, PCA, DBSCAN, stacked RoBERTa+GloVe embeddings), CombinedTM, Top2Vec, Llama2 with BERTopic, and Mixtral 8x7b — on the Cisco Labs software vulnerability dataset (69,909 entries, 39 columns) targeting the *Threat* feature.
- Applied UMAP and PCA for dimensionality reduction with HDBSCAN and DBSCAN clustering; CombinedTM aligned Bag-of-Words with SBERT contextual embeddings, while Llama2-7B-chat generated interpretable topic labels via prompt-guided labeling.
- Evaluated models on topic coherence and clustering quality; Mixtral 8x7b identified five contextually relevant topics per document without clustering or iterative training, demonstrating LLM-driven topic extraction as a scalable alternative to traditional pipeline-based approaches.

Addressing Unresolved Crimes in India: AI-Driven Insights for Effective Policymaking | *Under Review* **2026**

- S Anirudh, G Angelina, M Spoorthi, CR Kavitha. Manuscript under review. Performs exploratory data analysis on 40,160 unresolved crime records across Indian cities, employing Falcon-7B-Instruct for automated crime categorization and spatiotemporal visualization for law enforcement policymaking.

Evaluating BERT-Based Transformer Models for Improved Phishing Detection Using NLP Techniques | *Under Review* **2026**

- S Anirudh, M Spoorthi, CR Kavitha. Manuscript under review. Benchmarks BERT, RoBERTa, DistilBERT, and XLNet on phishing email classification with a four-strategy augmentation pipeline; RoBERTa achieved the highest accuracy (96.65%) and F1-score (96.64%).

KannaDa – A Kannada-based Programming Language Interpreter using Lex and Yacc | *IEEE*
IEEE Xplore **Jun. 2025**

- M Spoorthi, P Manaswini, S Shaik, CR Kavitha. Published in IEEE, June 2025. Develops a Kannada-syntax programming language interpreter using compiler design principles and NLP techniques to democratize coding for native Kannada speakers.

Advancing Supply Chain Management: Cloud-Based Demand Forecasting Solutions | *IEEE*
DOI: 10.1109/ICONAT61936.2024.10775167 **Dec. 2024**

- M Spoorthi, SS Rao, S Mishra, SM Rajagopal, M Prashanth. Published in IEEE ICONAT 2024. Explores AWS-based demand forecasting for supply chain management using AWS Forecast and ML algorithms for scalable inventory optimization.

Harnessing the Power of Web3: A Blockchain Approach to Crowdfunding | *IEEE*
IEEE Xplore **Jul. 2024**

- RV Savant, SN Sunder, M Spoorthi, SM Rajagopal, RM Balakrishnan. Published in IEEE, July 2024. Proposes a decentralized crowdfunding application using Ethereum, Solidity smart contracts, and MetaMask wallets to enable secure, transparent peer-to-peer fundraising.

A Comparative Analysis of Black-Box and White-Box Models for IoT Botnet Detection | *IEEE*

DOI: 10.1109/i2ct61223.2024.10544270 **May 2024**

- RV Savant, M Spoorthi, PK Mohan, S Bhaskaran, SM Rajagopal. Published in IEEE I2CT 2024. Employs a two-stage binary and multi-class ML classification pipeline on CTU-13 and N-BaIoT datasets; LGBM and XGBoost demonstrated superior botnet detection performance.

Unveiling Hidden Patterns: Clustering Algorithms on C Code Embedding | *IEEE*

IEEE Xplore **May 2024**

- M Spoorthi, RV Savant, S Seshadri, N Narmada, PB Pati. Published in IEEE, May 2024. Applies ML clustering (K-means, GMMs) to CodeBERT embeddings of C programs to uncover structural patterns, highlighting the limits of embedding-based automated code grading.

Additional Projects

Self-Supervised Pretraining for Brain Tumor Classification | *Ongoing* **Jun. 2025 – Present**

Collaboration with BITS Pilani Dubai

- Developing a contrastive self-supervised pretraining pipeline (SimCLR) on 29,459 brain MRI slices from NY, AZ, and UCSF cohorts for downstream GBM and metastasis classification.
- Achieved 89.90% accuracy and AUROC 0.9706 on held-out test set; fine-tuned on labeled downstream task following Stage 1 contrastive pretraining on unlabeled multi-institutional data.
- Running full SLURM-based training pipeline on Sharanga HPC with PyTorch, currently extending to Prima VLM for NIfTI-native 3D GBM classification.

Vision-Based Framework for Autonomous Navigation in Agricultural Domains |

Amrita Vishwa Vidyapeetham **Nov. 2024 – Mar. 2025**

- Designed a fully vision-based autonomous navigation framework for agricultural and warehouse environments using ZED X stereo cameras, vSLAM, and the Nav2 framework, eliminating dependency on LiDAR sensors.
- Integrated stereo vision, IMU fusion, GNSS localization, waypoint navigation, and recovery fallback mechanisms for robust real-time localization, 3D mapping, obstacle avoidance, and adaptive path planning in unstructured terrains.
- Extended the framework to support crop monitoring, sprayer integration, and human detection for smart farming applications.

Real-Time Pothole Detection for Day and Night Navigation using YOLO | *Ongoing* **Jan. 2026 – Present**

- Benchmarked YOLOv8 through YOLOv12 (15 variants across nano, small, and medium scales) for real-time pothole detection under daytime and nighttime conditions, using a custom multi-condition dataset collected across varying times of day.
- Best daytime performance achieved by YOLOv8s (mAP50 = 0.941, P = 0.939, R = 0.853); best nighttime performance by YOLOv10s (mAP50 = 0.522, P = 0.571); best overall cross-condition performance by YOLOv10s (mAP50 = 0.805, mAP50-95 = 0.457).
- Identified significant performance degradation under low-light conditions across all variants (daytime mAP50 up to 0.941 vs. nighttime up to 0.522), motivating future work on domain-adaptive and illumination-robust detection pipelines.
- Trained and evaluated all models on Google Colab cloud GPUs; results support deployment recommendations for autonomous navigation systems requiring reliable pothole detection across lighting conditions.

Experience

Accenture — **Oct. 2025 – Present**

Advanced Application Engineering Analyst — *Bengaluru, India*

- Maintain and optimize CI/CD pipelines in Azure DevOps, automating build, test, and deployment workflows across cloud environments on Microsoft Azure and AWS.
- Author smoke testing scripts in Pester (PowerShell) to validate post-deployment service health, and manage source control and release automation via GitHub Actions.

GE Aerospace — **Jan. 2025 – Jun. 2025**

Digital Technology Intern — *Bengaluru, India*

- Built and optimized RESTful APIs using Java (Spring Boot) with JPA and PostgreSQL for backend services, ensuring quality through JUnit, Postman, and code reviews.
- Containerized microservices with Docker for deployment via AWS CodePipeline.

Centre for Development of Telematics (C-DOT), Telecom R&D Institute, Govt. of India — **Jun. 2024 – Jul. 2024**

Internship Trainee – Network Management — *Bengaluru, India*

- Developed a comprehensive web application for real-time network monitoring, performance analysis, and failure reporting, dynamically retrieving and visualizing data from a MySQL database.
- Built modules for task execution log tracking, system status monitoring, failure rate analysis, and root cause identification using Java EE, JDBC, AngularJS, and MySQL.

Centre for Development of Telematics (C-DOT), Telecom R&D Institute, Govt. of India — **Jul. 2023 – Aug. 2023**

Internship Trainee – Data Science for 5G O-RAN — *Bengaluru, India*

- Developed machine learning solutions for 5G O-RAN use cases including context-based dynamic handover management for V2X communications, QoE/QoS optimization, local indoor positioning on RAN, and energy saving using Python and ML libraries.

Technical Skills

Languages: Python, Java, SQL, C++
Frameworks & Libraries: PyTorch, TensorFlow, OpenCV, HuggingFace, Transformers, PySpark
Tools & Platforms: AWS, Docker, Git, Linux, Hadoop, PostgreSQL, MySQL

Leadership / Extracurricular

National Student Data Corp (NSDC) – Bengaluru Chapter — **Jan. 2024 – Aug. 2025**

Co-Founder — *Amrita Vishwa Vidyapeetham, Bengaluru*

- Co-founded the Bengaluru chapter of the National Student Data Corp to foster a data science and machine learning research culture among students at Amrita School of Computing.
- Built a research ideation platform focused on AI, ML, and data science, enabling students to propose ideas, connect with peers sharing overlapping research interests, and form collaborative project teams.
- Coordinated faculty and mentor assignments for student-led AI/ML projects, providing end-to-end support from ideation through execution, with several projects progressing to conference and journal publication.

Forum for Aspiring Computer Engineers (FACE Amrita) — **Jan. 2023 – Jan. 2024**

Senior Executive — *Amrita Vishwa Vidyapeetham, Bengaluru*

- Organized hackathons, technical workshops, and tutorial sessions across various branches of computer science, fostering peer learning and practical skill development within the department.